

Edge-Guided Depth Completion Network

エッジ情報を利用したデプス補完ネットワーク

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Abstract

Depth completion refers to the process of creating dense, pixel-level depth maps from sparse data collected by sensors such as LiDAR (Light Detection and Ranging). This technology is extensively utilized in fields like autonomous driving and 3D mapping. While LiDAR provides precise distance measurements, the sparsity of its output presents challenges for tasks requiring dense depth information. Recent deep learning-based methods have made substantial progress in addressing these challenges, demonstrating improved accuracy and robustness. This paper offers a comprehensive review of depth completion, emphasizing benchmark datasets, evaluation metrics, architectural innovations, and learning strategies. Additionally, it identifies critical challenges, such as maintaining smooth transitions and sharp boundaries in sparse data, and proposes directions for future advancements to meet the demands of high-precision applications. This paper also offers a new method for the depth completion by using the edge information from the images to get a high quality of the completion of depth network.

1 Introduction

Depth information plays a vital role in tasks such as scene understanding, autonomous driving, and SLAM [1, 2, 3]. It is also essential for applications like augmented reality and precision agriculture, which require high accuracy in depth estimation [4, 5]. Although LiDAR provides accurate distance measurements [6], the resulting depth maps are typically sparse and inadequate for many real-world applications.

Depth completion aims to generate dense depth maps from sparse inputs to compensate for this limitation. However, the task faces two major challenges: extremely high sparsity often exceeding 95% and the need to adapt to diverse sensors and scenarios. In recent years, deep learning-based approaches have shown remarkable potential in tackling these challenges [7]. Despite significant progress achieved by deep learning models, capturing fine-grained geometric details remains a challenge. Therefore, incorporating geometric priors or auxiliary networks to enhance geometric representation has become increasingly important.

For instance, Binary Anisotropic Diffusion Tensor (B-ADT) [8] employs binary diffusion weights and an Image-Guided Nearest Neighbor Search (IGNNS) strategy to ensure smooth depth transitions within objects while preserving sharp disconti-

nities at boundaries, effectively enhancing the representation of geometric structures. Various convolutional neural network (CNN)-based models have been proposed in recent years. SparseConvNet [9] introduced sparse convolution to effectively process irregular input data. Convolutional Spatial Propagation Network (CSPN) [10] leveraged dynamic affinity matrices to guide depth propagation, achieving a balance between smooth transitions and sharp boundaries.

Furthermore, BP-Net [11] further advanced depth completion by utilizing bilateral propagation modules and multi-scale architectures, ensuring both structural consistency and local detail preservation. The latest state-of-the-art model, Distilling Monocular Foundation Model for Fine-grained Depth Completion [12], combines monocular foundation models with depth completion networks. By using geometric cues, this approach significantly enhances the accuracy of depth prediction, especially in preserving fine structural details.

The KITTI dataset [9] serves as a key benchmark for outdoor depth completion. However, its depth annotations are extremely sparse — as shown in Figure 1, valid depth measurements cover only about 5% of the pixels, primarily on foreground objects. Our objective is to complete these sparse measurements into a dense depth map with a pixel-wise density comparable to RGB images. Even in regions with relatively dense measurements, depth values at the boundaries between different objects are often missing, as illustrated in Figure 2. Preserving clear and accurate object boundaries during completion is critical for maintaining structural consistency and supporting downstream tasks. Therefore, recovering missing depth along object boundaries remains a significant challenge. Another notable issue in KITTI is the large-scale absence of depth in distant areas, especially in the upper part of many images, as shown in Figure 3. Accurately inferring depth in these regions requires leveraging geometric cues from RGB images and the available depth in nearby areas. Ensuring structural completeness in 3D space under such sparsity is an essential goal of robust depth completion methods.

This paper proposes a novel edge-guided multimodal depth completion network that integrates RGB images, sparse depth maps, and panoptic segmentation maps to enhance structural consistency and boundary sharpness. The main contributions of this study are as follows:

1. We leverage a pretrained Mask2Former model to extract high-quality object boundary information, which guides depth prediction and significantly improves completion per-

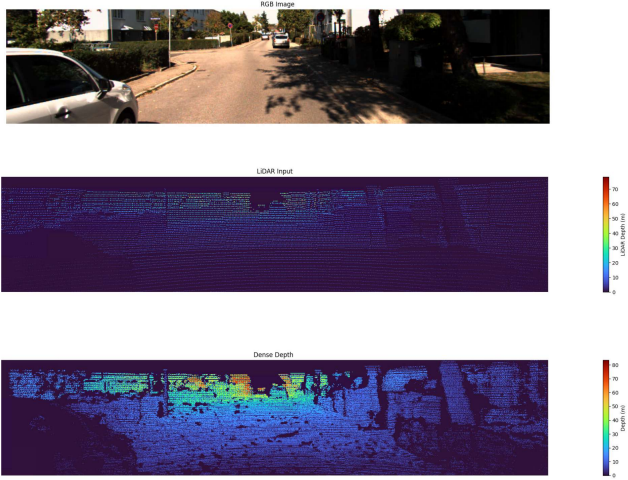


Fig. 1: Example of RGB image, sparse depth map, and the ground-truth dense depth map from the dataset.



Fig. 2: Illustration of missing object boundaries in the sparse depth map. Depth values around foreground objects such as the cyclist and pedestrian are absent.

formance at semantic boundaries, particularly in regions with severe sparse point missing.

2. We design a three-stage depth completion framework consisting of preprocessing, multimodal fusion, and post-processing. This structure enables the progressive integration of semantic, geometric, and visual features, significantly improving structural completeness and depth map clarity.
3. In the preprocessing stage, we introduce a region-aware propagation strategy that selects the nearest valid sparse points only within the same semantic region, avoiding cross-object propagation and ensuring intra-object consistency.

Extensive experiments on the KITTI dataset demonstrate that the proposed model excels in preserving boundary details and reconstructing distant structures. Although the RMSE does not yet surpass state-of-the-art methods, our approach shows clear advantages in structural representation and perceptual consistency.

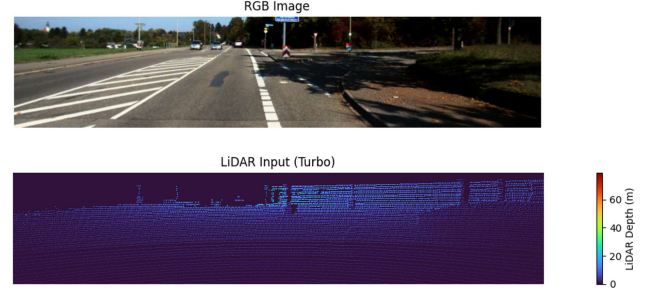


Fig. 3: Demonstration of missing depth in distant regions (e.g., the sky). The upper-left part of the image lacks valid depth values, highlighting the severe sparsity of LiDAR data in far-field and upper areas.

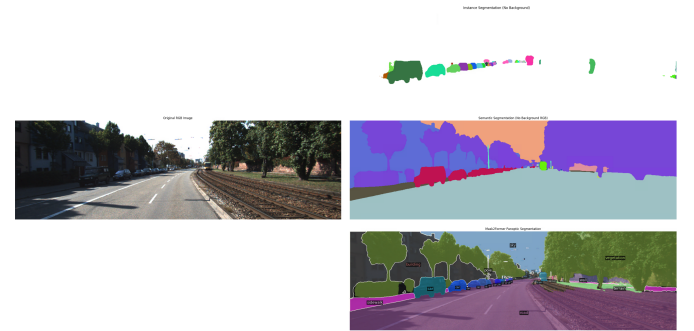


Fig. 4: Visualization of three types of segmentation: panoptic, instance, and semantic segmentation. These methods provide structural guidance for depth completion [13]

2 Related work

2.1 Extraction and utilization of object boundaries in images

In depth completion and image understanding tasks, object boundary information plays a crucial role. To extract high-quality boundaries, image segmentation techniques are commonly used. The three most typical segmentation types are semantic segmentation, instance segmentation, and panoptic segmentation.

Image segmentation is a pixel-level visual processing technique that aims to identify and group semantically consistent regions within an image. According to whether the target is countable, image elements are generally classified into two categories: *things* and *stuff*. *Things* refer to countable objects such as people, cars, and animals, while *stuff* represents uncountable regions like sky, grass, and roads.

Semantic segmentation focuses on uncountable "stuff", assigning a class label to each pixel. For example, all pixels belonging to the sky are labeled as "sky", without distinguishing between different sky regions.

Instance segmentation focuses on countable "things", not

only recognizing the category of objects but also distinguishing between individual instances, such as labeling two cars as "car 1" and "car 2".

Panoptic segmentation combines the strengths of both semantic and instance segmentation. Each pixel is assigned both a semantic label and an instance ID, providing a unified representation of "stuff" and "things".

These segmentation methods are particularly important in real-world applications like autonomous driving. For instance, 3D semantic segmentation can identify various scene elements on the road, helping vehicles better understand their surroundings. Meanwhile, instance segmentation enables identification of individual objects, which supports depth estimation and path planning [14].

Although these tasks differ semantically, many existing methods are designed for only one type of segmentation. For example, semantic segmentation often uses pixel-wise classification based on fully convolutional networks (FCNs). Recently, the *Masked-Attention Mask Transformer (Mask2Former)* has been proposed as a unified solution for all segmentation tasks. It consists of a backbone feature extractor, a pixel decoder, and a Transformer decoder, and can be used for semantic, instance, and panoptic segmentation tasks [13].

Therefore, in this work, we adopt the panoptic segmentation architecture of Mask2Former and utilize its pretrained model to extract high-quality object boundary information, which is further used in our depth completion framework.

2.2 Boundary-Aware Fusion of Semantic and Depth Features

As proposed in the PanDepth model [15], combining RGB images, semantic features, and sparse depth maps in joint panoptic segmentation and depth completion can effectively improve both depth prediction accuracy and semantic consistency. Specifically, the model first concatenates the RGB image with the initial semantic segmentation map and feeds them into two 2D convolutional layers to extract joint semantic-visual features. The semantic map is generated by the EfficientPS [16] network, which is a top-down panoptic segmentation method that assigns a semantic label (e.g., car, road, tree) to each pixel and an instance ID to pixels belonging to "thing" categories, enabling fine-grained pixel-level understanding.

At the same time, the sparse depth map is processed separately by 2D convolutions and then fused with the RGB-semantic features through a set of Fuse Blocks. In these blocks, the model uses 2D convolutions to capture image-plane features, while continuous convolutions are applied to extract spatial structure from the sparse depth data, allowing 2D and 3D features to be effectively combined. The fused features are then refined by two more convolutional layers to produce a high-quality dense depth map. Notably, this depth map is not only one of the model's outputs but is also passed back into the semantic segmentation branch via a Joint Branch, where it serves as guidance to improve the sharpness and consistency of

semantic boundaries.

2.3 Depth completion network:the Bilateral Propagation Network BP-Net

To address the problem of incomplete information and blurry object boundaries in sparse depth maps, Tang et al. [11] proposed the Bilateral Propagation Network (BP-Net). This model integrates three key modules—bilateral propagation, multi-modal fusion, and depth refinement—into a unified multi-scale framework for jointly modeling image appearance and depth information. BP-Net adopts a symmetric multi-scale structure where depth estimation and refinement are performed progressively across different resolutions.

In the preprocessing stage, BP-Net fuses the sparse depth map and RGB image to generate an initial dense depth map. Specifically, the target pixel depth D'_i is computed as a weighted sum over N neighboring valid sparse points:

$$D'_i = \sum_{j \in \mathcal{N}(i)} \omega_{ij} D_{ij}^c = \sum_{j \in \mathcal{N}(i)} \omega_{ij} (\alpha_{ij} S_j + \beta_{ij}) \quad (1)$$

Here, α_{ij} , β_{ij} , and ω_{ij} denote affine transformation coefficients and combination weights. These are generated by a multilayer perceptron (MLP), conditioned on the image encodings $\mathcal{I}_i, \mathcal{I}_j$, the source depth encoding $\mathcal{S}_j$, and spatial offset encoding $\mathcal{O}_{ij}$. The neighborhood $\mathcal{N}(i)$ is determined using Euclidean distance in the image plane, and empirically the top 4 nearest valid pixels ($N = 4$) are used for propagation. The propagation process relies on jointly learning from image, depth, and geometric cues.

At each resolution scale, the initial dense depth is back-projected to the camera space to extract depth features, which are concatenated with image features and fed into a U-Net-based multi-modal fusion module. This module consists of:

- **Encoder:** ResNet blocks extract hierarchical image features;
- **Decoder:** Deconvolution and skip connections fuse low- and high-level features;

Output is a residual depth map is predicted and added to the previous depth to form the updated depth D''_s . To further improve the quality of object boundaries, a CSPN++ [17]-like refinement module is incorporated at each scale. It uses image-guided local linear propagation to iteratively refine the depth map and alleviate boundary ambiguity.

BP-Net extends conventional one or two-stage pipelines by introducing an explicit preprocessing stage, forming a three-stage framework that improves prediction accuracy and robustness.

The entire network is trained in an end-to-end manner across six resolution scales (Scale 0 to 5) with a weighted multi-scale L2 loss:

$$\mathcal{L} = \sum_{s=0}^5 \sum_{i \in \mathcal{P}_v} \lambda_s \|D_i^{gt} - \mathcal{U}_s(D_s)_i\|^2 \quad (2)$$

where $\mathcal{P}_v$ is the set of valid ground-truth depth pixels, $\mathcal{U}_s$ denotes bilinear upsampling from scale s to the original resolution, and λ_s is the loss weight at each scale.

Experiments conducted on both indoor (NYUv2) and outdoor (KITTI) benchmarks show that BP-Net achieves state-of-the-art performance, ranking first on the official KITTI depth completion leaderboard at submission time. These results confirm the effectiveness of the bilateral propagation mechanism in handling sparse inputs, as well as the benefits of multi-modal fusion and refinement in producing sharp and consistent depth boundaries.

2.4 Depth completion network: Distilling Monocular Foundation Model for Fine-grained Depth Completion DMD-3C

To address the issue of incomplete predictions and blurry boundaries caused by relying solely on sparse ground truth and L1 loss, Liang et al. proposed a two-stage knowledge distillation framework — *Distilling Monocular Foundation Model for Fine-grained Depth Completion* [12]. This framework fully leverages the geometric priors encoded in monocular depth estimation models and enhances depth prediction accuracy and structural consistency through a carefully designed loss formulation.

The backbone model used in this distillation framework is the previously introduced BP-Net [11], which incorporates multi-scale propagation, image-guided fusion, and refinement modules. It efficiently extracts joint features from sparse depth and image content, offering strong representational capacity for knowledge transfer.

In the first stage, the monocular depth model f_{mo} is used to generate pseudo depth maps from unlabeled RGB images $\mathcal{I}_{un}$:

$$D_{syn} = f_{mo}(\mathcal{I}_{un}) \quad (3)$$

Synthetic sparse depth maps D_{simu} are then generated via simulated scanning and combined with the RGB images as input to the BP-Net:

$$D_f = f_{\theta}^{BP}(\mathcal{I}_{un}, D_{simu}) \quad (4)$$

The predicted dense depth D_f is supervised by the pseudo label D_{syn} using an L1 loss:

$$\mathcal{L}_{pre} = \|D_f - D_{syn}\|_1 \quad (5)$$

This pretraining stage enables the model to learn global structures and spatial priors embedded in the monocular model, thereby improving the initial prediction capability.

In the second stage, the BP-Net [11] remains the backbone, but the training objective shifts from mimicking monocular pseudo depth to reinforcing structural consistency under real-world sparse supervision. Specifically, the input now includes labeled RGB images $\mathcal{I}_{label}$ and their corresponding sparse depth maps D_{sp} . The network predicts a completed depth map D_f , which is first supervised by the sparse ground truth and then aligned with the monocular prediction D_m , thereby enabling structural refinement. BP-Net leverages its multi-scale propagation mechanism to facilitate structural feature transfer and employs edge-aware strategies to enhance accuracy near depth discontinuities.

Given the inherent scale and shift ambiguity in monocular predictions, direct supervision using D_m can be misleading. To resolve this, a Scale- and Shift-Invariant Loss (SSI Loss) is introduced to align D_f with D_m while allowing for global normalization:

$$\mathcal{L}_{SSIL} = \min_{s,b} \|D_f - (s \cdot D_m + b)\|_1 \quad (6)$$

This loss effectively normalizes any global differences in scale and offset. By minimizing the loss across all image pixels, the model produces depth maps that are consistent with the relative structures captured by the monocular model.

To further improve prediction sharpness and alignment at depth discontinuities, a gradient matching regularization term is added:

$$\mathcal{L}_{reg} = \frac{1}{N} \sum_{k=1}^K (|\nabla_x R_k| + |\nabla_y R_k|), \quad R = D_f - D_m \quad (7)$$

Here, $K = 4$ denotes the number of scale levels, each obtained by halving the resolution. R_k represents the residual depth difference at the k -th scale.

The final training objective integrates all three components:

$$\mathcal{L}_{total} = \mathcal{L}_{sup} + \mathcal{L}_{SSIL} + \mathcal{L}_{reg} \quad (8)$$

where $\mathcal{L}_{sup}$ is the standard supervised loss computed from sparse ground truth.

Experimental results demonstrate that this framework, which combines strong monocular priors with the expressive BP-Net architecture, achieves state-of-the-art performance on the KITTI dataset, particularly excelling in structural fidelity and edge sharpness.

2.5 Limitations of Existing Methods

Although existing depth completion methods such as BP-Net and DMD3C have achieved promising results in recovering the overall scene structure, they still suffer from noticeable limitations in terms of boundary sharpness and depth estimation in distant regions.

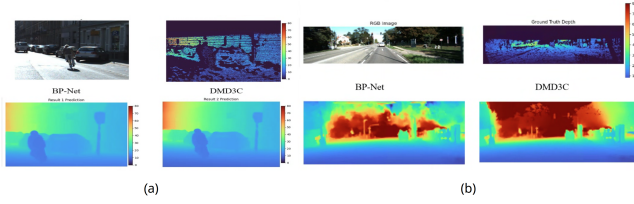


Fig. 5: The comparison of creation results

As illustrated in the visual comparisons in Fig. 5, both BP-Net and DMD3C show blurring artifacts in depth prediction near object edges. While the overall geometric structure is reasonably restored, the completion results around foreground objects, such as pedestrians and traffic lights remain inaccurate. In particular, vertical structures like lamp posts exhibit ambiguous separation boundaries, leading to unclear transitions between objects. This degrades the quality of scene understanding and hinders accurate 3D reconstruction.

In the case of distant regions, BP-Net often fails to produce valid depth responses, resulting in large blank or missing areas. Although DMD3C improves coverage and completes more pixels in far-field regions, it still performs poorly in preserving structural continuity and fine details in the upper parts of the image. For example, object contours between trees and buildings are often fragmented and inconsistent with the geometric boundaries seen in the RGB image, indicating limited modeling capacity under highly sparse input conditions.

3 Proposed method

3.1 Problem Definition

Mainstream depth sensors such as LiDAR produce highly sparse depth maps. For instance, in the KITTI dataset, only about 5% of the pixels contain valid depth values, which are primarily located on foreground objects. Depth values in distant regions and along object boundaries are often missing or incomplete.

The core objective of depth completion is to reconstruct a dense depth map with a resolution comparable to the RGB image by leveraging sparse depth inputs along with auxiliary information such as RGB images or semantic segmentation maps. This task presents several key challenges: how to accurately reconstruct depth structures at varying distances under severely sparse input conditions; how to preserve depth continuity and geometric consistency in boundary regions and avoid blurring or misalignment. To address these issues, it is essential to design depth completion models that are structure-aware and capable of fusing multi-scale information effectively.



Fig. 6: Panoptic segmentation of the dataset



Fig. 7: Estimation of Depth Anything V2 on the dataset

3.2 Network architecture

3.2.1 Data Preparation

We first generate the panoptic segmentation maps and corresponding dense depth maps to guide the depth completion process. The segmentation maps are produced using a Mask2Former model pretrained on the Cityscapes dataset, which contains 8 “thing” classes and 11 “stuff” classes. As shown in Fig. 6, this model can effectively distinguish semantic regions such as roads, pedestrians, vehicles, and trees. Since depth discontinuities often occur along object boundaries, extracting such semantic edge information is crucial for improving structural consistency and boundary sharpness in depth completion. For dense depth maps, we adopt the Depth Anything V2 model to perform monocular depth estimation on natural images, yielding aligned dense depth predictions, as shown in Fig. 7. These maps reflect the scene’s depth structure and foreground-background relationships. The final grayscale depth

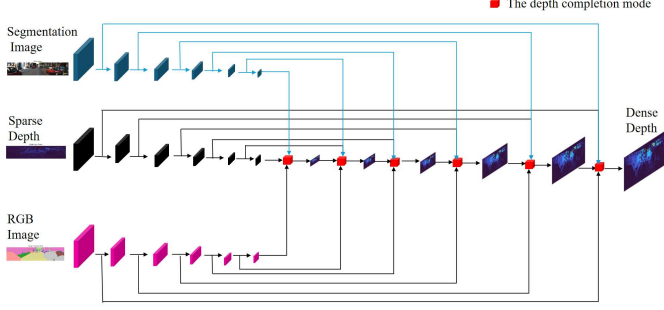


Fig. 8: The architecture of the New network

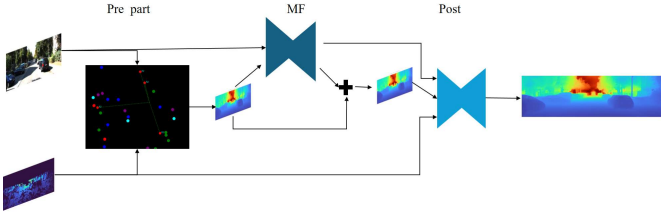


Fig. 9: New depth completion part

maps, along with the segmentation maps, are used as inputs for training the network.

3.2.2 Network Architecture

We propose a multimodal depth completion architecture as shown in Fig. 8 that integrates semantic segmentation maps, RGB images, and sparse depth inputs. The network takes three types of input: segmentation images, sparse depth maps, and RGB images. Each input branch passes through its own encoder to extract semantic, geometric, and appearance features, respectively. These features are progressively fused in the middle layers via a sequence of fusion modules. In each fusion module (highlighted in red in the figure), the image features, geometric features (from sparse depth), and semantic features (from the segmentation map) at the current layer are fused to predict an intermediate depth map. This intermediate depth map not only serves as the output of the current stage but also acts as the input to the next stage, enabling further refinement of the depth prediction.

To further enhance the feature integration, the central part of the network employs a three-stage fusion design consisting of a Pre-processing part (Pre), a Multi-modal Fusion part (MF), and a Post-refinement part (Post). As shown in Fig. 9 In the Pre stage, we use a pixel propagation strategy guided by both RGB and semantic information to convert sparse depth into an initial dense depth map. Unlike conventional KNN-based methods, our approach selects the four nearest valid pixels within the same semantic region to ensure intra-object consistency and

prevent cross-object propagation. The initial depth map is then combined with RGB features and fed into the MF module, which uses an encoder-decoder structure to learn a residual depth map. This residual is added to the initial estimate to produce an intermediate depth output. The intermediate result is subsequently passed to the Post module, where CSPN++ is used for image-guided local propagation and refinement. A multi-scale strategy is also adopted to improve edge detail and structural consistency at different resolutions. In future work, we plan to replace the Post module with a more flexible fusion mechanism to further improve performance.

Eventually, the fusion modules produce a dense depth map with high boundary sharpness and strong structural coherence.

Loss Function. Depending on whether monocular depth supervision is used, we employ two different loss strategies. In the default setting without monocular guidance, the network is optimized solely by the sparse ground truth supervision loss $\mathcal{L}_{\text{sup}}$, as defined in Eq.2. When monocular depth prediction (e.g., from Depth Anything V2) is available, we introduce use Eq.8

This combined loss enables the model to benefit from both sparse supervision and monocular priors, improving its ability to recover scene structure and preserve edge sharpness.

4 Experiments

4.1 Dataset:KITTI Depth Completion Dataset

The KITTI depth completion dataset is a well-known and extensively used resource for outdoor research. It comprises over 93,000 semi-dense depth maps, along with corresponding sparse LiDAR scans and RGB image pairs. The dataset is structured into 86,000 training samples, 7,000 validation samples, and 1,000 for testing. With a resolution of 1216×352 , it supports applications that require high-resolution data. [18].

4.2 Evaluation Metrics

Depth completion and monocular depth estimation generally share the same evaluation metrics. We list the most commonly used measures as follows [7]:

Root Mean Squared Error (RMSE) is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}, \quad (9)$$

where Y_i and $\hat{Y}_i$ are the ground truth and predicted depth values, respectively, and n is the total number of pixels.

The above metrics are commonly used in depth completion and monocular depth estimation tasks. Specifically, RMSE is frequently adopted for evaluation in the KITTI benchmark, which ranks algorithms based on RMSE values. Consequently, many methods have used RMSE as the loss function for training models.

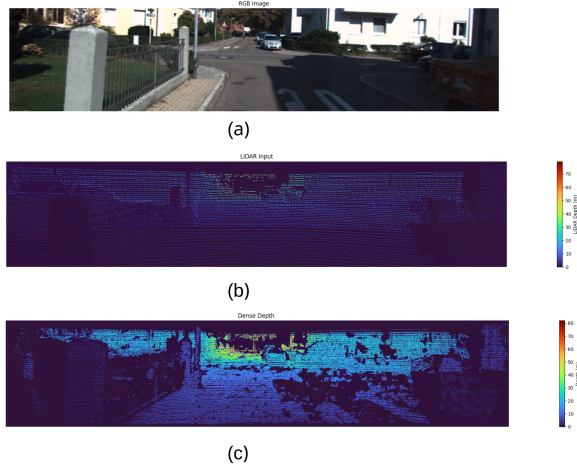


Fig. 10: Sample images from the KITTI depth completion dataset [19]. (a) RGB images. (b) Raw sparse depth maps. (c) Ground truth depth maps.[19].

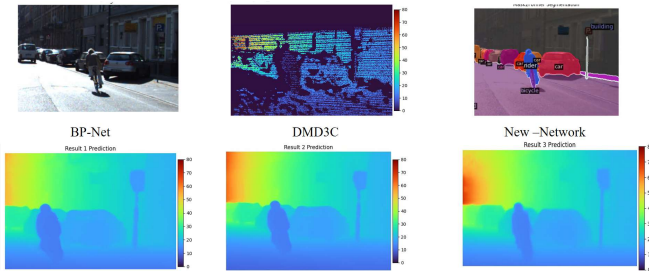


Fig. 11: Comparison of edge prediction results across different depth completion networks

4.3 Experiment results

As shown in Fig. 11, the depth map predicted by **BP-Net** appears relatively smooth, capturing the general outlines of objects such as pedestrians and vehicles. However, the edges tend to be blurry, and boundaries between adjacent objects are sometimes indistinct. While BP-Net demonstrates strong capabilities in overall structural restoration, it still lacks precision in preserving fine-grained details and boundary sharpness. In contrast, **DMD3C** produces depth predictions with better edge clarity, where object contours particularly for pedestrians and foreground elements are more prominent, indicating higher sen-

Table 1: RMSE comparison of different methods.

Method	RMSE
BP-Net	0.718
DMD-3C	0.715
New Network	0.802

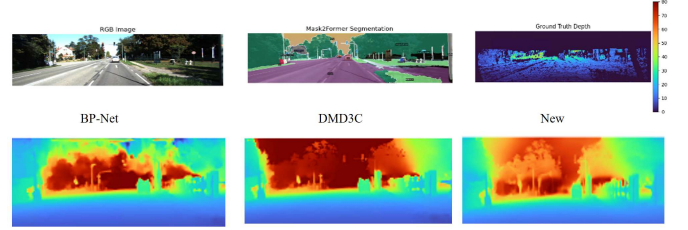


Fig. 12: Comparison of depth prediction performance in distant regions across different completion networks

sitivity to depth discontinuities and edge regions. Furthermore, the proposed **New Network**, enhanced with panoptic segmentation information, significantly improves the perceptibility of structural boundaries.

As depicted in Fig. 12, in distant regions such as the sky and horizon, BP-Net fails to generate valid depth responses, resulting in noticeable holes and missing regions. This reflects the model’s limited capability in handling extremely sparse inputs over long-range areas. Compared to BP-Net, DMD3C can generate relatively complete depth predictions for distant areas (e.g., sky, trees), achieving better spatial coverage, although it still lacks smoothness and accuracy in fine structures and depth transitions. The **New Network**, which integrates semantic guidance, exhibits superior performance in modeling distant regions. It not only ensures coverage of far-field areas but also recovers structural details of buildings and trees with higher fidelity, making its predictions more consistent with real-world geometry.

Nevertheless, it is worth noting that in terms of the RMSE metric, the **New Network** has not yet surpassed the performance of existing models, indicating room for improvement in controlling global prediction error. In future work, replacing the current generation module or introducing more refined loss strategies could help further improve its accuracy and stability.

5 Conclusion

This paper systematically investigates the key challenges and development directions in the task of depth completion. We propose a novel multi-modal depth completion network that integrates semantic segmentation information. The network effectively combines RGB images, sparse depth maps, and panoptic segmentation maps generated by Mask2Former. Through a three-stage architecture: preprocessing, multi-modal fusion, and depth refinement—it progressively enhances the structural consistency and edge quality of the predicted depth maps.

In the experiments, we compare the proposed New Network with two baselines: the classic BP-Net and the two-stage knowledge distillation model DMD-3C. As shown in Fig. 11, the New Network leverages semantic priors to significantly improve object contour clarity, especially at the boundaries of foreground

targets such as pedestrians and vehicles. Regarding long-range depth estimation, Fig. 12 demonstrates that the New Network is better able to reconstruct distant depth structures like the sky, trees, and buildings, effectively addressing the modeling limitations of previous methods in ultra-sparse regions.

However, it is worth noting that in terms of RMSE metrics, the New Network does not yet outperform BP-Net and DMD-3C, indicating a trade-off between numerical accuracy and edge detail preservation.

Future work could focus on two aspects:

(1) Designing a more flexible post-processing module (e.g., replacing CSPN++ with an improved propagation strategy).

(2) Introducing more robust joint supervision mechanisms to enhance the model’s adaptability to complex scenes and long-range structures.

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